

2.5.3. Product of Maxterms

Each of the 2^n functions of n binary variables can also be expressed as a product of maxterms. To express the Boolean function as a product of maxterms, it must first be brought into a form of OR terms. This may be done by using the distributive law $[x+yz = (x+y)(x+z)]$. Then any missing variable x in each OR term is ORed with x' .

Example 2-6: Express the Boolean function $F = xy + x'z$ in a product of maxterm form.

$$\begin{aligned}
 F = xy + x'z &= (x+y)(x+z) \quad (\text{Postulate 4(b)}) \\
 &= (x'+x)(x+y)(x+z) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+x) \cdot (x+y)) \cdot ((x+z) \cdot (x+y)) \quad (\text{Postulate 4(b)}) \\
 &= ((x'+x') \cdot (x'+y)) \cdot ((x+z) \cdot (x+y)) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+x') \cdot (x'+y)) \cdot ((x+z) \cdot (x+y)) \quad (\text{Postulate 3(a)}) \\
 &= (1 \cdot (x'+y)) \cdot ((x+z) \cdot (x+y)) \quad (\text{Postulate 5(a)}) \\
 &= (x'+y) \cdot 1 \cdot ((x+z) \cdot (x+y)) \quad (\text{Postulate 3(b)}) \\
 &= (x'+y) \cdot ((x+z) \cdot (x+y)) \quad (\text{Postulate 2(b)}) \\
 &= (x'+y+0) \cdot (((x+z)+0) \cdot ((x+y)+0)) \quad (\text{Postulate 2(a)}) \\
 &= ((x'+y)+zz') \cdot (((x+z)+yy') \cdot ((x+y)+x'x')) \quad (\text{Postulate 5(b)}) \\
 &= ((x'+y+z) \cdot (x'+y+z')) \cdot (((x+z+y) \cdot (x+z+y')) \cdot ((x+y+z) \cdot (x+y+z'))) \\
 &= ((x'+y+z) \cdot (x'+y+z')) \cdot (((x+z+y) \cdot (x+z+y')) \cdot ((x+y+z) \cdot (x+y+z'))) \quad (\text{Postulate 4(b)}) \\
 &= ((x'+y+z) \cdot (x'+y+z')) \cdot (((x+z+y) \cdot (x+z+y')) \cdot ((x+y+z) \cdot (x+y+z'))) \quad (\text{Postulate 3(b)}) \\
 &= ((x'+y+z) \cdot (x'+y+z')) \cdot (((x+z+y) \cdot (x+z+y')) \cdot ((x+y+z) \cdot (x+y+z'))) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+y+z) \cdot (x'+y+z')) \cdot (((x+z+y) \cdot (x+z+y')) \cdot ((x+y+z) \cdot (x+y+z'))) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+y+z) \cdot (x'+y+z')) \cdot (((x+z+y) \cdot (x+z+y')) \cdot ((x+y+z) \cdot (x+y+z'))) \quad (\text{Theorem 4(a)}) \\
 &= ((x'+y+z) \cdot (x'+y+z')) \cdot (((x+z+y) \cdot (x+z+y')) \cdot ((x+y+z) \cdot (x+y+z'))) \quad (\text{Postulate 3(a)})
 \end{aligned}$$